



Foundations of Programming

Assignment No. : 13

PROBLEM STATEMENT:

Write a program in C to accept a number from the user and print the digits of the number in reverse order.

OBJECTIVE:

- To understand number manipulation using loops
- To learn the use of modulo (%) and division (/) operators.
- To implement logic for reversing digits of a number.
- To improve problem-solving skills using iterative structures.

THEORY:

- To understand number manipulation using loops.
- To learn the use of modulo (%) and division (/) operators.
- To implement logic for reversing digits of a number.
- To improve problem-solving skills using iterative structures.

PLATFORM:

Linux

ALGORITHM:

C PROGRAM:

```
#include<stdio.h>
```

```
int main()
```

```
{
```

```
    int n, rev=0, rem;
```

```
printf("Enter a number: ");  
  
scanf("%d",&n);  
  
while(n!=0)  
{  
    rem = n%10;  
    rev = rev*10 + rem;  
    n = n/10;  
}  
  
printf("Reversed number = %d",rev);  
}
```

INPUT:

```
Enter a number: 12345
```

OUTPUT:

```
Reversed number = 54321
```

CONCLUSION:

Thus, the program to print the digits of a number in reverse order was successfully implemented using C programming. The program demonstrates the effective use of loops, modulo operator, and division operator.

FAQs:

1. Why is the modulo operator (%) used in this program?
2. Why is division by 10 used after extracting each digit?

3. What happens if the input number is zero?
4. How will the program behave for negative numbers?
5. Why is a while loop preferred for this operation?